

CHE 201 Spring 2020 HW 5

Anna Christensen

TOTAL POINTS

92.5 / 100

QUESTION 1

1 Question 1 15 / 15

- ✓ - **0 pts** Correct
 - **5 pts** wrong specific volume
 - **5 pts** wrong T2
 - **5 pts** wrong T when the tank contains only saturated vapor
 - **15 pts** Question not answered

QUESTION 2

2 Question 2 10 / 15

- **0 pts** Correct
- ✓ - **5 pts** wrong v1
 - **5 pts** wrong v2
 - **5 pts** wrong work
 - **15 pts** Question not answered

QUESTION 3

3 Question 3 15 / 15

- ✓ - **0 pts** Correct
 - **5 pts** Wrong T-v diagram
 - **2.5 pts** wrong U1
 - **2.5 pts** wrong u2
 - **2.5 pts** wrong electricity work
 - **2.5 pts** wrong time
 - **15 pts** Question not answered

QUESTION 4

4 Question 4 20 / 20

- ✓ - **0 pts** Correct
 - **20 pts** Question not answered
 - **5 pts** wrong u1
 - **5 pts** wrong u2
 - **5 pts** wrong electricity work
 - **5 pts** wrong final heat transfer

QUESTION 5

5 Question 5 20 / 20

- ✓ - **0 pts** Correct
 - **5 pts** wrong sketch of p-v diagram
 - **2.5 pts** wrong Q1
 - **2.5 pts** wrong W1
 - **2.5 pts** wrong Q2
 - **2.5 pts** wrong W2
 - **2.5 pts** wrong Q3
 - **2.5 pts** wrong W3
 - **20 pts** Question not answered

QUESTION 6

6 Question 6 12.5 / 15

- **0 pts** Correct
- ✓ - **2.5 pts** not enough components of ways to improve water use
 - **5 pts** not enough components of ways to improve water use
 - **7.5 pts** not enough components of ways to improve water use
 - **2.5 pts** not enough components of record of water use and compare
 - **5 pts** not enough components of record of water use and compare
 - **15 pts** Question not answered

HW 5

2/23/2020

3.18

①

$$T_1 = -20^\circ\text{C} \quad T_2 = ?$$

$$P_2 = 6 \text{ bar}$$

$$V = \text{constant}$$

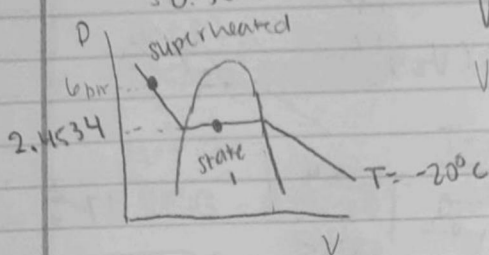
$$V_2 = V_1$$

$$x = 50.36\%$$

$$= 0.5036$$

$$V_g = 0.0926$$

$$V_f = 0.7427 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$$



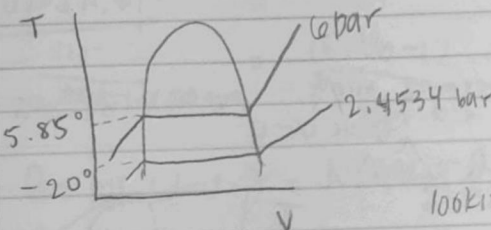
$$V_2 = (1-x)V_f + xV_g$$

$$V_2 = (1-0.5036)(0.7427 \times 10^{-3})$$

$$+ 0.5036(0.0926)$$

$$V_2 = 0.047 \frac{\text{m}^3}{\text{kg}}$$

$$P_2 = 6 \text{ bar}$$



looking at table A-9

at 6 bar w/ $0.047 \frac{\text{m}^3}{\text{kg}}$ as V

$$T_2 = 45^\circ\text{C}$$

The temperature at which the tank contains only saturated vapor = 0°C

3.24

②

0.5 lb water

$$P_2 = 14.7 \frac{\text{lb}_f}{\text{in}^2}$$

$$PV^{1.2} = \text{constant}$$

$$P_1 = 40 \frac{\text{lb}_f}{\text{in}^2}$$

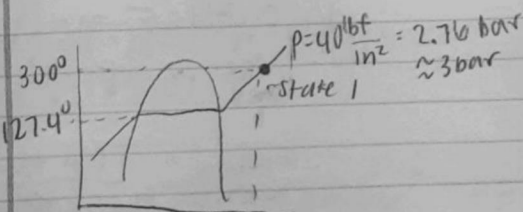
$$W = ?$$

$$W = \int P dv$$

$$V_1 = ?$$

$$V_2 = ?$$

$$T_1 = 300^\circ\text{F}$$



$$P = 3 \text{ bar}$$

$$T = 280 \rightarrow V = 0.844$$

$$T = 320 \rightarrow V = 0.907$$

$$\frac{320-280}{0.907-0.844} = \frac{300-280}{V_1-0.844}$$

$$V_1 = 0.8755$$

1 Question 1 15 / 15

✓ - 0 pts Correct

- 5 pts wrong specific volume
- 5 pts wrong T2
- 5 pts wrong T when the tank contains only saturated vapor
- 15 pts Question not answered

HW 5

2/23/2020

3.18

①

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$$P_2 = 6 \text{ bar}$$

$$V = \text{constant}$$

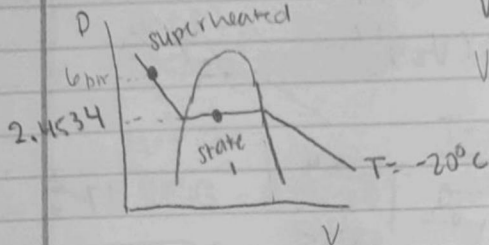
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$$x = 50.36\%$$

$$= 0.5036$$

$$V_g = 0.0926$$

$$V_f = 0.7427 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$$



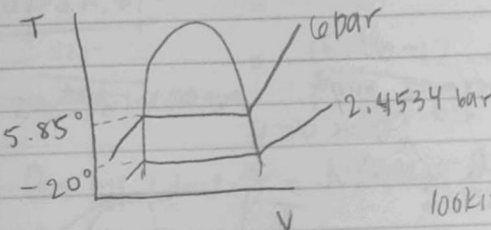
$$V_2 = (1-x)V_f + xV_g$$

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$$+ 0.5036(0.0926)$$

$$V_2 = 0.047 \frac{\text{m}^3}{\text{kg}}$$

$$P_2 = 6 \text{ bar}$$



looking at table A-9

at 6 bar w/ $0.047 \frac{\text{m}^3}{\text{kg}}$ as V

$$T_2 = 45^\circ\text{C}$$

The temperature at which the tank contains only saturated vapor = 0°C

3.24

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0.5 lb water

$$P_2 = 14.7 \frac{\text{lb}_f}{\text{in}^2}$$

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$$P_1 = 40 \frac{\text{lb}_f}{\text{in}^2}$$

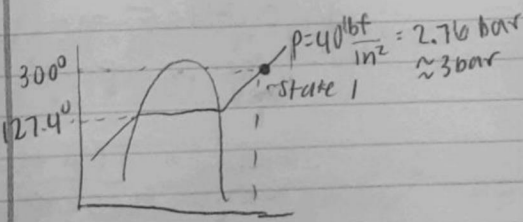
$$W = ?$$

$$W = \int P dv$$

$$V_1 = ?$$

$$V_2 = ?$$

$$T_1 = 300^\circ\text{F}$$



$$P = 3 \text{ bar}$$

$$T = 280 \rightarrow V = 0.844$$

$$T = 320 \rightarrow V = 0.907$$

$$\frac{320-280}{0.907-0.844} = \frac{300-280}{V_1-0.844}$$

$$V_1 = 0.8755$$

$$P_1 V_1^{1.2} = P_2 V_2^{1.2}$$

$$P_1 = 40 \frac{\text{lb}_f}{\text{in}^2} = 2.76 \text{ bar}$$

$$V_1 = 0.8755$$

$$P_2 = 14.7 \frac{\text{lb}_f}{\text{in}^2} = 1.01 \text{ bar}$$

$$(2.76)(0.8755)^{1.2} = (1.01)(V_2)^{1.2}$$

$$V_2 = 1.884 \frac{\text{m}^3}{\text{kg}}$$

$$W = \int_{V_1}^{V_2} \frac{1}{V^{1.2}} dV = -\frac{5}{V^{0.2}} \Big|_{0.8755}^{1.884} = 0.7297 \text{ J}$$

$$= 6.92 \times 10^{-4} \text{ Btu}$$

③ 1.5 kg

2 kW

determine time

$v = \text{constant}$

$$x = 0.6$$

$$= 2 \frac{\text{kJ}}{\text{s}}$$

$$D \ll \Delta V = 0$$

$$T_1 = 0^\circ \text{C}$$

$$Q - W = \Delta U$$

$$Q = m(u_2 - u_1)$$

$$u_1 = (1-x)u_f + x u_g$$

$$= (1-0.6)(49.79) + 0.6(227.06)$$

$$u_1 = 156.152 \frac{\text{kJ}}{\text{kg}}$$

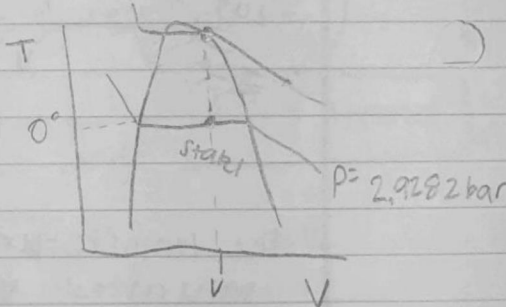
$$V_f = 0.7721 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$$

$$V_g = 0.0689$$

$$V = (1-x)V_f + x V_g$$

$$= (1-0.6)(0.7721 \times 10^{-3}) + 0.6(0.0689)$$

$$V = 0.04165 = V_2 = V_g$$



2 Question 2 10 / 15

- 0 pts Correct

✓ - 5 pts wrong v1

- 5 pts wrong v2

- 5 pts wrong work

- 15 pts Question not answered

$$P_1 V_1^{1.2} = P_2 V_2^{1.2}$$

$$P_1 = 40 \frac{\text{lb}_f}{\text{in}^2} = 2.76 \text{ bar}$$

$$V_1 = 0.8755$$

$$P_2 = 14.7 \frac{\text{lb}_f}{\text{in}^2} = 1.01 \text{ bar}$$

$$(2.76)(0.8755)^{1.2} = (1.01)(V_2)^{1.2}$$

$$V_2 = 1.884 \frac{\text{m}^3}{\text{kg}}$$

$$W = \int_{V_1}^{V_2} \frac{1}{V^{1.2}} dV = -\frac{5}{V^{0.2}} \Big|_{0.8755}^{1.884} = 0.7297 \text{ J}$$

$$= 6.92 \times 10^{-4} \text{ Btu}$$

③ 1.5 kg

$$x = 0.6$$

$$T_1 = 0^\circ \text{C}$$

$$2 \text{ kW}$$

$$= 2 \frac{\text{kJ}}{\text{s}}$$

determine time

$v = \text{constant}$

$$Q - W = \Delta U$$

$$Q = m(u_2 - u_1)$$

$$u_1 = (1-x)u_f + x u_g$$

$$= (1-0.6)(49.79) + 0.6(227.06)$$

$$u_1 = 156.152 \frac{\text{kJ}}{\text{kg}}$$

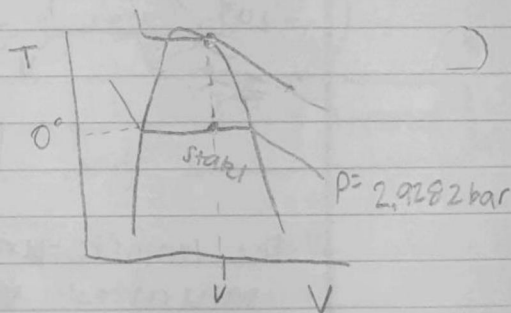
$$V_f = 0.7721 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$$

$$V_g = 0.0689$$

$$V = (1-x)V_f + x V_g$$

$$= (1-0.6)(0.7721 \times 10^{-3}) + 0.6(0.0689)$$

$$V = 0.04165 = V_2 = V_g$$



$$T = 12^\circ\text{C} \rightarrow u_g = 233.63 \quad v_g = 0.0460$$

$$T = 16^\circ\text{C} \rightarrow u_g = 235.78 \quad v_g = 0.04165 \text{ between } 12^\circ \text{ and } 16^\circ$$

$$\rightarrow v_g = 0.0405$$

$$\frac{16-12}{235.78-233.63}$$

$$v = \frac{16-12}{0.0405-0.0460} = \frac{T-12}{0.04165-0.0460}$$

$$T = 15.16^\circ$$

$$\frac{16-12}{235.78-233.63} = \frac{15.16-12}{x-233.63} \rightarrow x = 235.3 \frac{\text{kJ}}{\text{kg}} = u_2$$

$$Q = m(u_2 - u_1) = 1.5 \text{ kg} (235.3 - 156.152 \frac{\text{kJ}}{\text{kg}})$$

$$Q = 118.7 \text{ kJ} \times \frac{1 \text{ s}}{2 \text{ kg}} = \boxed{59.4 \text{ seconds}}$$

3.44 (4) $T_1 = -10^\circ\text{C}$ 12V 5 amp 5min $V_2 = V_1$
 $x = 0.8$ $T_2 = 40^\circ\text{C}$
 $V = 0.01 \text{ m}^3$

at -10°C :

$$u_f = 33.27$$

$$u_g = 223.02$$

$$u = (1-x)u_f + x u_g$$

$$u = (0.2)(33.27) + 0.8(223.02)$$

$$u_f = 185.07 \frac{\text{kJ}}{\text{kg}}$$

$$v_f = 0.7606 \times 10^{-3}$$

$$v_g = 0.0652 \frac{\text{m}^3}{\text{kg}}$$

$$v = 0.2(0.7606 \times 10^{-3}) + 0.8(0.0652)$$

$$v_1 = v_2 = 0.0523 \frac{\text{m}^3}{\text{kg}}$$

at 40°C : where $v = 0.0523$

$$u = 250.2 \frac{\text{kJ}}{\text{kg}}$$

3 Question 3 15 / 15

✓ - 0 pts Correct

- 5 pts Wrong T-v diagram
- 2.5 pts wrong U_1
- 2.5 pts wrong u_2
- 2.5 pts wrong electricity work
- 2.5 pts wrong time
- 15 pts Question not answered

$$\begin{aligned}
 T = 12^\circ\text{C} &\rightarrow u_g = 233.63 & v_g = 0.0460 \\
 T = 16^\circ\text{C} &\rightarrow u_g = 235.78 & v = 0.04165 \text{ between } 12^\circ \text{ and } 16^\circ \\
 & & v_g = 0.0405
 \end{aligned}$$

$$\frac{16-12}{235.78-233.63}$$

$$v = \frac{16-12}{0.0405-0.0460} = \frac{T-12}{0.04165-0.0460}$$

$$T = 15.16^\circ$$

$$\frac{16-12}{235.78-233.63} = \frac{15.16-12}{x-233.63} \rightarrow x = 235.3 \frac{\text{kJ}}{\text{kg}} = u_2$$

$$Q = m(u_2 - u_1) = 1.5 \text{ kg} (235.3 - 156.152 \frac{\text{kJ}}{\text{kg}})$$

$$Q = 118.7 \text{ kJ} \times \frac{1 \text{ s}}{2 \text{ kg}} = \boxed{59.4 \text{ seconds}}$$

3.44 (4) $T_1 = -10^\circ\text{C}$ 12V 5 amp 5min $V_2 = V_1$
 $x = 0.8$ $T_2 = 40^\circ\text{C}$
 $V = 0.01 \text{ m}^3$

$$\text{at } -10^\circ\text{C}:$$

$$u_f = 33.27$$

$$u_g = 223.02$$

$$u = (1-x)u_f + x u_g$$

$$u = (0.2)(33.27) + 0.8(223.02)$$

$$u_f = 185.07 \frac{\text{kJ}}{\text{kg}}$$

$$v_f = 0.7606 \times 10^{-3}$$

$$v_g = 0.0652 \frac{\text{m}^3}{\text{kg}}$$

$$v = 0.2(0.7606 \times 10^{-3}) + 0.8(0.0652)$$

$$v_1 = v_2 = 0.0523 \frac{\text{m}^3}{\text{kg}}$$

$$\text{at } 40^\circ\text{C}: \text{ where } v = 0.0523$$

$$u = 250.2 \frac{\text{kJ}}{\text{kg}}$$

$$\text{mass} = \frac{V}{v} = \frac{0.01 \text{ m}^3}{0.0523 \frac{\text{m}^3}{\text{kg}}} = 0.1912 \text{ kg}$$

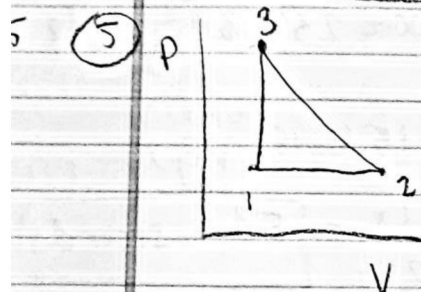
$$W = -V_i \Delta t = -(12 \text{ V} \times 5 \text{ A} \times 5 \text{ min} \times \frac{60 \text{ sec}}{1 \text{ min}} \times \frac{1 \text{ kJ}}{10^3 \text{ J}})$$

$$= -18 \text{ kJ}$$

$$Q = U + W$$

$$= m(u_2 - u_1) + W = 0.1912(250.2 - 185.07) - 18$$

$$Q = -5.51 \text{ kJ}$$



process 1-2

$$\frac{T_1}{V_1} = \frac{T_2}{V_2}$$

$$T_2 = \frac{1.4 V_1}{V_2} (500)$$

$$T_2 = 700^\circ \text{R}$$

$$W = \int p dV = p(V_2 - V_1) \quad pAV = nRT$$

$$= nR(T_2 - T_1) = 1$$

$$116 = 0.454 \text{ kg}$$

$$^{\circ}\text{R} = -272.594^\circ \text{C}$$

$$= (0.454 \text{ kg})(0.08206)(115.759 - 4.6778)$$

$$= (116)(0.069)(700 - 500) = 13.716 \text{ ft}^2$$

$$W_{12} = 13.71 \text{ Btu}$$

$$Q_{12} = U_{12} + W_{12}$$

$$U_{12} = m(u_2 - u_1)$$

$$\text{at } T = 500^\circ \text{R}$$

$$u_1 = 85.2 \text{ Btu/lb}$$

$$\text{at } T = 700^\circ \text{R}$$

$$u_2 = 119.58 \text{ Btu/lb}$$

$$Q_{12} = (116)(119.58 - 85.2) + 13.71 = 48.09 \text{ Btu} = Q_{12}$$

Process 2-3

$$Q_{23} = 0 \quad W = -m(u_3 - u_2)$$

$$W = -(1)(140.47 - 119.58)$$

$$\text{at } T = 20^\circ \text{R}$$

$$u_3 = 140.78 \text{ Btu/lb}$$

$$= -20.89 \text{ Btu} = W_{23}$$

4 Question 4 20 / 20

✓ - 0 pts Correct

- 20 pts Question not answered
- 5 pts wrong u1
- 5 pts wrong u2
- 5 pts wrong electricity work
- 5 pts wrong final heat transfer

$$\text{mass} = \frac{V}{v} = \frac{0.01 \text{ m}^3}{0.0523 \frac{\text{m}^3}{\text{kg}}} = 0.1912 \text{ kg}$$

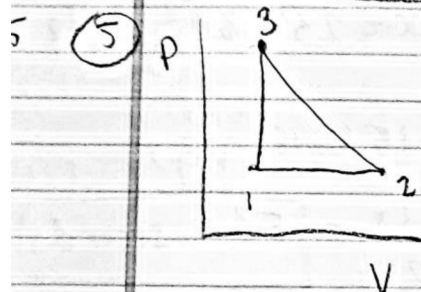
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process 1-2

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$$Q_{12} = (116)(119.58 - 85.2) + 13.71 = 48.09 \text{ Btu} = Q_{12}$$

Process 2-3

$$Q_{23} = 0 \quad W = -m(u_3 - u_2)$$

$$W = -(1)(140.47 - 119.58)$$

$$\text{at } T = 20^\circ \text{R}$$

$$u_3 = 140.78 \text{ Btu/lb}$$

$$= -20.89 \text{ Btu} = W_{23}$$

process 3-1

$$W_{31} = 0$$

$$Q = m(u_1 - u_3) = (1116)(85.71 - 140.47)$$

$$= -55.27 \text{ Btu} \approx Q_{31}$$

net work

$$W = 13.17 + (-20.89) + 0 = -7.18 \text{ Btu}$$

net heat transfer

$$Q = 48.09 + 0 + (-55.27) = -7.18 \text{ Btu}$$

6)

During a typical day, I drink about 2 liters a water, which is about half of a gallon, which is the recommend amount of water by health authorities.

(<https://www.healthline.com/nutrition/how-much-water-should-you-drink-per-day>)

Most people I know shower at least once a day, and the typical shower lasts around 8.2 minutes, with a flowrate of 2.1 gallons per minute, which uses 17.2 gallons of water. Showers are the 3rd largest water use in homes after toilets and clothes washers. A way we could reduce our water usage is by switching shower head, which is a relatively cheap option, as well as to reduce the time we spend in the shower. (<https://www.home-water-works.org/indoor-use/showers>)

5 Question 5 20 / 20

✓ - 0 pts Correct

- 5 pts wrong sketch of p-v diagram
- 2.5 pts wrong Q1
- 2.5 pts wrong W1
- 2.5 pts wrong Q2
- 2.5 pts wrong W2
- 2.5 pts wrong Q3
- 2.5 pts wrong W3
- 20 pts Question not answered

process 3-1

$$W_{31} = 0$$

$$Q = m(u_1 - u_3) = (1116)(85.71 - 140.47)$$

$$= -55.27 \text{ Btu} \approx Q_{31}$$

net work

$$W = 13.17 + (-20.89) + 0 = -7.18 \text{ Btu}$$

net heat transfer

$$Q = 48.09 + 0 + (-55.27) = -7.18 \text{ Btu}$$

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In older toilets, the average flush uses about 3.6 gallons, which adds up to 18.8 gallons a day per person. The ultra-low-flow (ULF) toilets have an average flush volume of 1.6 gallons, which adds up to 9.1 gallons a day. In 1992, Federal law ordered that all toilets made in the US must use an average of 1.6 gallons. There are also high efficiency toilets (HET) which flush at a maximum of 1.3 gallons, as well as dual-flush toilets, which have half-flush and full flush options, which have an average flush of 1.1 gallons. (<https://www.home-water-works.org/indoor-use/toilets>)

There is a system that continually keeps hot water flowing through pipes, so that no water is wasted while someone is waiting for it to warm up. This hot water on demand system does use up a lot more energy than a typical system since it is running constantly. Also, this option is very expensive. It is expensive to heat water, and as the water goes through the pipes it loses heat, which creates more of an expense. There are multiple types of this system, including ones where timers can be adjusted and temperature controlled. Overall I believe the expense of it will not encourage people to use this option to try and save water.

(<https://chilipepperapp.com/index.php/fast-hot-water-saves-water/>)

In offices, 50 liters of water are used each day per employee. A way to combat water usage in offices would be to ensure that installers of new plumbing systems are aware of the requirements of the Water Regulations enforcement. Another would be to let staff know of the intention to save water. Hose pipes and sprinklers can use 1000 liters per hour, so keeping grass watering to a minimum. Collecting rain water is a good way to find water to use for water plants. (<https://www.south-staffs-water.co.uk/media/1509/waterusebusiness.pdf>)

6 Question 6 12.5 / 15

- 0 pts Correct
- ✓ - 2.5 pts not enough components of ways to improve water use
- 5 pts not enough components of ways to improve water use
- 7.5 pts not enough components of ways to improve water use
- 2.5 pts not enough components of record of water use and compare
- 5 pts not enough components of record of water use and compare
- 15 pts Question not answered